



S Class Messages

S-Class train describer messages provide updates relating to the status of signalling equipment within a train describer area. Not all TD areas provide this information.

Message types

There are three types of S-Class message:

Type	Name	Details
SF	Signalling Update	Updates the signalling data for a specified set of signalling elements
SG	Signalling Refresh	Part of a set of SG messages which update the whole set of signalling data for TD area. Terminated by an SH message.
SH	Signalling Refresh Finished	Last message in a signalling refresh

S-Class messages report the status of the following signalling elements:

- **Signal aspects** - whether the signal is displaying its most restrictive aspect or not
- **Points** - whether the points are set normal or reverse
- **Route set** - the route set from a signal to another signal or bay platform
- **TRTS buttons** - the status of a Train Ready to Start button, used by train dispatch staff to indicate to the signaller that departure is due.
- **Level crossings** - whether they are lowered or raised
- **Track sections** - whether a particular track section (track circuit or axle counter section) is occupied or not

Not all train describers support S-Class data, and those that do may only support a subset of the available groups of data. The List of Train Describers shows S-Class support. You should also note that even though a subset may available it might not be complete. (E.g A TD may send SIG & TRK data but it may only relate to some signals and track sections - not all of the physical ones in a TD area)

Message Format

Messages are sent in JSON format, as follows:

```
{"SF_MSG":{"time":"1422404915000","area_id":"SI","address":"16","msg_type":"SF","data":"43"}}
{"SG_MSG":{"time":"1422404915000","area_id":"RW","address":"00","msg_type":"SG","data":"06880306"}}
{"SH_MSG":{"time":"1422404915000","area_id":"RW","address":"04","msg_type":"SH","data":"35000000"}}
```

Message Fields

Field	SF	SG	SH	Details
time	✓	✓	✓	Message time UNIX timestamp <i>in milliseconds</i> since the UNIX epoch
area_id	✓	✓	✓	Alphanumeric code representing the TD area that the message originates from (see list of train describers)
msg_type	✓	✓	✓	Type of message Can be SF_MSG, SG_MSG or SH_MSG
address	✓	✓	✓	Address of the signalling data being updated. For an SF_MSG, this is the individual signalling element. For an SG_MSG or SH_MSG, this is the starting address for the four bytes supplied in the data field.
data	✓	✓	✓	Signalling data

Working with S-Class data

In order to use the S-Class data contained in these messages you need to know what each element is. Network Rail have agreed to release a number of technical documents for each TD which will provide the information you require. While this process has begun it will take some time to complete - you should check the [list of train describers](#) for any information that may already have been released or user contributions. You may also like to have the challenge of trying to decode it yourself, in which case you should read the section below and the article on [Decoding S-Class Data](#).

- ECS Specifications - This provides the full external communications specification showing all connections to and from a TD along with the S-Class bit maps required. Where multiple connections are listed data to the NROD platform generally come via the 'SMART' connection.
- SOP Tables - These are the signalling output tables for a TD and show all bytes and bits

IECCs have ECSs and most other TDs have SOPs.

For meanings of individual signalling functions, see [Signalling Nomenclature](#).

Interpreting S-Class data

S-Class data represents the state of individual signalling elements within a TD area as bits within a set of up to 2,048 bits. The number of bits needed for a TD area depends on the number of signalling elements within that area. Each time an element changes state, an SF message is sent with the value of the elements in the relevant byte.

Every few hours, the state of all bits is sent as a 'refresh' in a series of SG messages with the value of groups of four bytes. These messages are terminated with an SH message.

To decode the set of addresses and bits in to signalling elements, a mapping table containing a list of bits and the signalling elements to which they refer is needed. To reference this to signals on the track layout, a diagram containing signal numbers, routes available from signals, point numbers

and track circuit detail is also needed. For advice on deducing your own mapping table, see [Decoding S-Class Data](#).

Network Rail Open Data Feeds	
Data Feeds	About the Feeds • Account States • Durable Subscriptions • Example Code (PHP / C# / Java / Ruby / Node.js) • STOMP Client Libraries • Advanced Uses • FAQ • Release Notes
RTPPM	RTPPM Feed
Train Movements	Train Movements Feed • Train Activation • Train Cancellation • Train Movement • Train Reinstatement • Change of Origin • Change of Identity • Change of Location • TSPEED Field • Planned Cancellations • Cancellation Codes
TD	TD Feed • C-Class Messages • S-Class Messages • Train Describers • TD Berths
VSTP	VSTP Feed
TSR	TSR Feed • Route Codes
SCHEDULE	SCHEDULE Feed • TIPLOC Records • Schedule and Location Records • Association Records • CIF Codes • How Scheduling Works • Allowances
Reference Data	Reference Data Feed • TOC Codes • CIF Codes • Delay Attribution Codes • Identifying Locations (STANOX, TIPLOC, NLC and 3-Alpha Codes) • STANOX Geographical Areas • Train Planning data (BPLAN)

Retrieved from "https://wiki.openraildata.com/index.php?title=S_Class_Messages&oldid=2726"